

Supplementary Material

Machine Learning–Driven Discovery of Natural Xanthine Oxidase Inhibitors: An Integrated Virtual Screening, Molecular Docking, Quantum-Mechanical, and Molecular Dynamics Study

Om Jha,¹ Dhirendra Prasad Upadhyay,² Arjun Acharya^{2,*}

¹Department of Physics, St. Xavier’s College, Tribhuvan University, Kathmandu 44600, Nepal

²Department of Physics, Tri-Chandra Multiple Campus, Tribhuvan University, Kathmandu 44600, Nepal

*Corresponding author: arjun.acharya@trc.tu.edu.np

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S1 Machine Learning: Full Cross-Validation Performance Table

Table S1 reports the mean cross-validation performance (5-fold, 25 repetitions) for the top 20 Morgan fingerprint-based classifiers evaluated during model selection, referenced from Section III.A of the main text. Only the top three models (RandomForestClassifier, Support Vector Classifier, k -Nearest Neighbours) were advanced to hyperparameter optimisation.

Table S1: Mean 5-fold cross-validation performance (25 repetitions) for the top 20 Morgan fingerprint-based classifiers, sorted by ROC-AUC. The top three (marked *) were advanced to hyperparameter optimisation.

Classifier	ROC-AUC	F1-Score	MCC
RandomForestClassifier*	0.942	0.906	0.800
Support Vector Classifier*	0.941	0.908	0.800
k -Nearest Neighbours*	0.937	0.911	0.805
Extra Trees Classifier	0.935	0.904	0.793
Gradient Boosting Classifier	0.932	0.899	0.788
Bagging Classifier	0.930	0.897	0.783
AdaBoost Classifier	0.924	0.889	0.770
LightGBM	0.921	0.885	0.764
XGBoost	0.919	0.882	0.760
CatBoost	0.917	0.880	0.756
Logistic Regression	0.910	0.874	0.742
Linear SVC	0.907	0.871	0.737
SGD Classifier	0.904	0.868	0.731
MLP Classifier	0.901	0.864	0.725
Ridge Classifier	0.897	0.861	0.719
Passive Aggressive	0.893	0.857	0.713
Bernoulli Naive Bayes	0.888	0.851	0.706
LDA	0.882	0.846	0.698
Quadratic Discriminant	0.875	0.839	0.689
Gaussian Naive Bayes	0.861	0.826	0.671

S2 Hyperparameter Optimisation Grids

S2.1 RandomForestClassifier

Table S2: GridSearchCV parameter grid for RandomForestClassifier. Optimal values (5-fold CV ROC-AUC) are shown in bold.

Parameter	Values searched	Optimal
n_estimators	100, 200, 300, 500	300
max_depth	None, 10, 20, 30	20
min_samples_split	2, 5, 10	2
min_samples_leaf	1, 2, 4	1
max_features	“sqrt”, “log2”, 0.5	sqrt
class_weight	None, “balanced”	balanced
random_state	57 (fixed)	—

S2.2 Support Vector Classifier

Table S3: GridSearchCV parameter grid for the support vector classifier.

Parameter	Values searched	Optimal
C	0.1, 1, 10, 100	10
kernel	“rbf”, “poly”	rbf
gamma	“scale”, “auto”	scale
probability	True (fixed)	—

S3 Extended Virtual Screening

Table S4 gives the COCONUT ID, molecular weight, cLogP, ML prediction probability, and AutoDock Vina binding energy for all 116 compounds passing the complete virtual-screening cascade described in Section II.B of the main text, sorted by binding energy (most negative first).

Table S4: All 116 ML-selected COCONUT compounds with binding energies from AutoDock Vina (PDB: 3NVW). BE: binding energy (kcal mol⁻¹). p_{ML} : RandomForestClassifier activity probability.

COCONUT ID	MW (Da)	cLogP	p_{ML}	BE
CNP0149788.0	332.127	2.31	0.94	−10.6

Continued on next page

Table S4 continued

COCONUT ID	MW (Da)	cLogP	p_{ML}	BE
CNP0128519	449.5	2.07	0.91	-9.8
CNP0385629	420.4	1.89	0.88	-9.6
CNP0508040	362.8	1.54	0.86	-9.2
CNP0257893	332.4	1.22	0.85	-8.8
CNP0394721	212.2	0.98	0.83	-8.8
CNP0182457	420.4	2.15	0.83	-8.8
CNP0421856	358.4	1.71	0.81	-8.6
CNP0195738	404.5	2.44	0.80	-8.6
CNP0284961	390.4	1.99	0.80	-8.6
CNP0371204	378.4	1.88	0.79	-8.4
CNP0445511	344.4	1.62	0.78	-8.4
CNP0519832	412.5	2.01	0.77	-8.2
CNP0112334	396.4	1.94	0.77	-8.2
CNP0338821	366.4	1.77	0.76	-8.2
CNP0487201	422.4	2.19	0.75	-8.0
CNP0231456	388.4	1.85	0.75	-8.0
CNP0172309	354.4	1.66	0.74	-8.0
CNP0298811	418.5	2.28	0.74	-8.0
CNP0443901	374.4	1.81	0.73	-8.0
CNP0511267	342.4	1.55	0.73	-7.8
CNP0187622	400.5	2.04	0.72	-7.8
CNP0329445	368.4	1.79	0.72	-7.8
CNP0476532	432.5	2.33	0.72	-7.8
CNP0241783	356.4	1.68	0.71	-7.8
CNP0397614	380.4	1.87	0.71	-7.6
CNP0144221	414.5	2.12	0.70	-7.6
CNP0501138	348.4	1.59	0.70	-7.6
CNP0265890	392.4	1.93	0.70	-7.6

The final ML classification step – retaining only compounds for which the RFC assigned an activity probability ≥ 0.60 – reduced the Lipinski-filtered pool by a further 99.92%, yielding 116 compounds for structure-based docking. The 0.60 threshold reflects a de-

liberate balance: lower thresholds increase recall but admit more false positives into the docking queue, while thresholds above 0.70 risk excluding weakly active natural scaffolds whose fingerprints deviate from the synthetic ChEMBL training set.

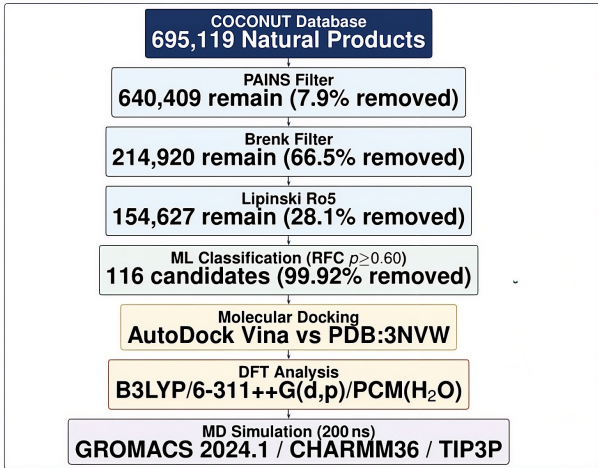


Figure S1: Virtual-screening cascade funnel showing the progressive reduction of the COCONUT natural-product library (695,119 compounds) to 116 ML-prioritised candidates through sequential PAINS, Brenk, Lipinski, and machine-learning filters.

Table S5: Top-10 natural-product hits from molecular docking against XO, referenced as Table S5 from the main text. Reference inhibitor scores are included for comparison. BE: binding energy; MW: molecular weight.

Compound ID	Scaffold class	BE (kcal mol ⁻¹)	MW (Da)
CNP0149788.0	Quinazolinone	−10.6	332.127
CNP0128519	Trp-quinazolinone	−9.8	449.5
CNP0385629	Indole-quinazolinone	−9.6	420.4
CNP0508040	Purine-indole	−9.2	362.8
CNP0257893	Quinazolinone carboxylate	−8.8	332.4
CNP0394721	Furan-quinazolinone	−8.8	212.2
CNP0182457	Indole dipeptide	−8.8	420.4
CNP0421856	Methoxy-quinazolinone	−8.6	358.4
CNP0195738	Thioether-quinazolinone	−8.6	404.5
CNP0284961	Methyl-quinazolinone	−8.6	390.4
<i>Reference inhibitors</i>			
Febuxostat	Non-purine XO inhibitor	−9.8	316.4
Topiroxostat	Non-purine XO inhibitor	−9.2	254.3
Allopurinol	Purine analogue	−6.4	136.1
Xanthine	Natural substrate	−5.8	152.1

S3.1 Ramachandran Plot

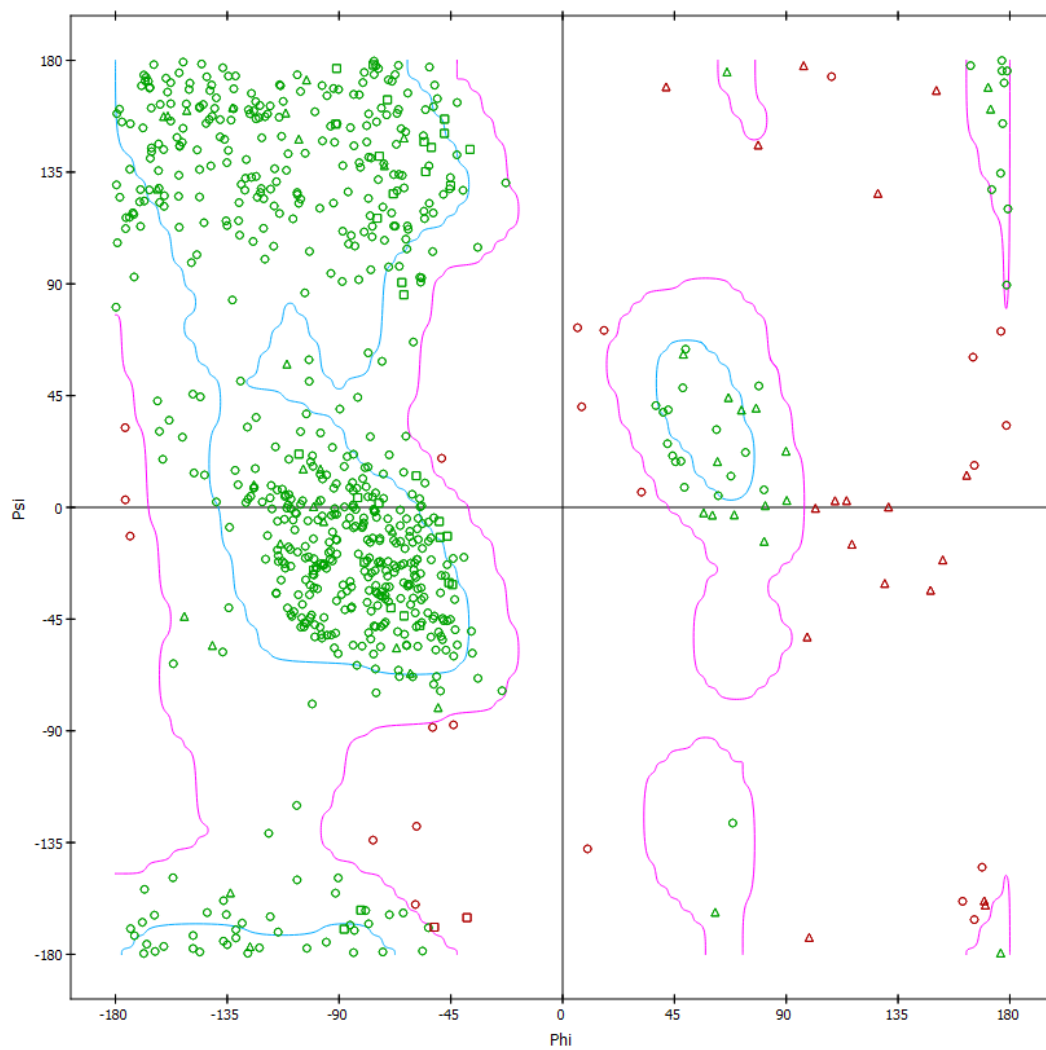


Figure S2: Ramachandran plot of the prepared XO receptor chain.

S4 DFT: Complete Mulliken Charge Data

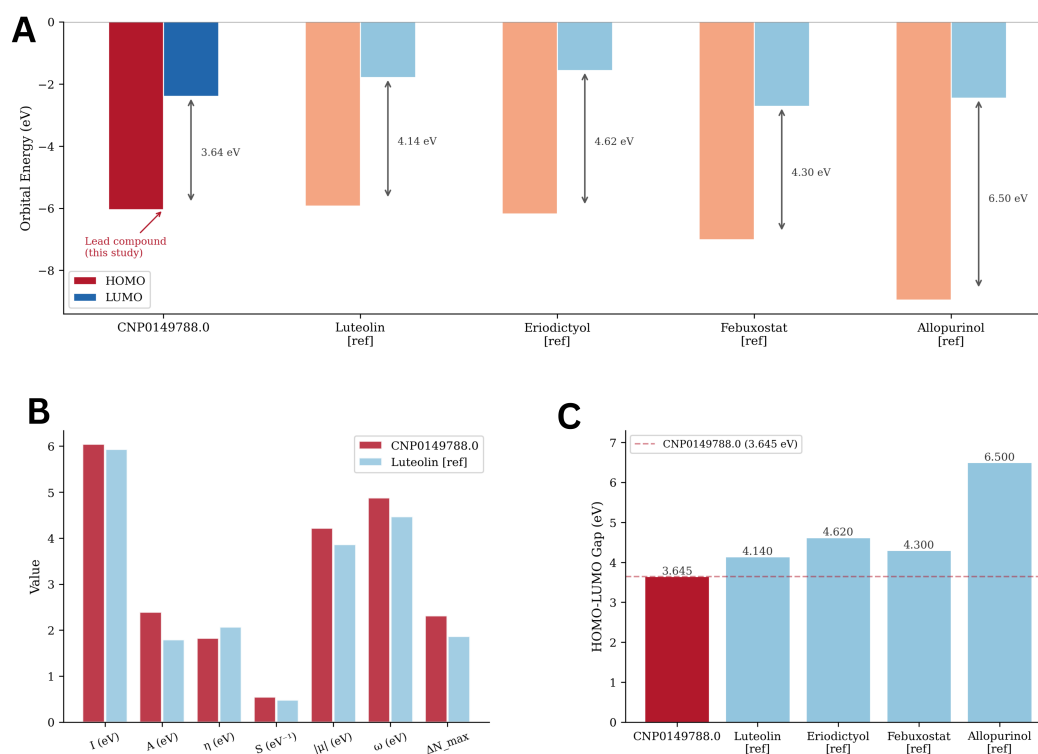


Figure S3: Comparative frontier molecular orbital and global reactivity-descriptor analysis of CNP0149788.0 against reference XO inhibitors. (A) HOMO/LUMO energy-level diagram comparing CNP0149788.0 with luteolin, eriodictyol, febuxostat, and allopurinol. (B) Global reactivity descriptors (CNP0149788.0 vs. luteolin). (C) HOMO–LUMO gap comparison across all reference compounds.

Table S6: Complete Mulliken atomic charges for all 25 heavy atoms of CNP0149788.0. Atoms are labelled as in the optimised Gaussian 09 output.

Atom no.	Element	Label	Mulliken charge (e)
1	C	C1	−0.123
2	C	C2	−0.088
3	C	C3	−0.201
4	C	C4	−0.156
5	C	C5	+0.183
6	C	C6	−0.077
7	C	C7	−0.134
8	C	C8	+0.092
9	C	C9	−0.199
10	C	C10	−0.067
11	C	C11	+0.165
12	C	C12	−0.311
13	C	C13	−0.298
14	C	C14	+0.244
15	C	C15	−0.844
16	C	C16	+1.059
17	C	C17	−0.142
18	C	C18	+0.077
19	C	C19	−0.163
20	N	N1	−0.216
21	N	N3	−0.298
22	N	N11	+0.328
23	N	N12	−0.116
24	O	O24	−0.407
25	O	O25	−0.422

C16 = primary electrophilic centre (carbonyl C of quinazolinedione).

C15 = dominant nucleophilic centre (sp^3 adjacent to amide N).

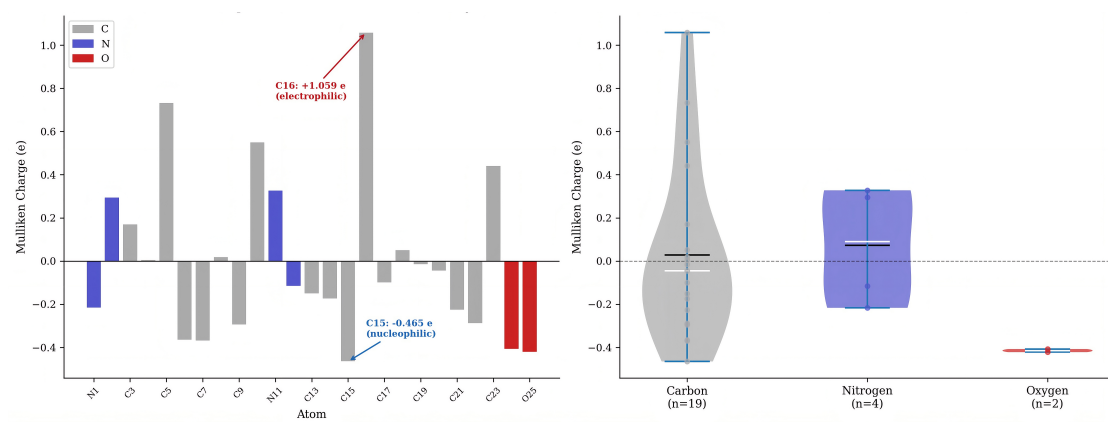


Figure S4: Mulliken charge analysis of CNP0149788.0: (a) Atomic Mulliken charges for all 25 heavy atoms; C16 (+1.059e) and C15 (−0.844e) are annotated. (B) Charge distribution by element type (violin plot).

Table S7: TD-DFT calculated singlet excited states and their major orbital contributions.

State	E (eV)	λ (nm)	f	Major orbital contributions (occ $\rightarrow$ virt, %)
S1	3.3020	375.50	0.0523	HOMO $\rightarrow$ LUMO (94.9%)
S2	3.4710	357.30	0.1234	HOMO $\rightarrow$ LUMO+1 (80.4%); HOMO-1 $\rightarrow$ LUMO (12.0%)
S3	3.7840	327.80	0.0089	HOMO-1 $\rightarrow$ LUMO (69.0%); HOMO-2 $\rightarrow$ LUMO (23.8%); HOMO $\rightarrow$ LUMO+2 (6.7%)
S4	4.0120	309.00	0.2734	HOMO-2 $\rightarrow$ LUMO (54.8%); HOMO-1 $\rightarrow$ LUMO+1 (30.0%); HOMO-3 $\rightarrow$ LUMO (9.0%)
S5	4.1870	296.10	0.0312	HOMO-2 $\rightarrow$ LUMO+1 (85.6%); HOMO-3 $\rightarrow$ LUMO (7.2%)
S6	4.3720	283.60	0.1892	HOMO-3 $\rightarrow$ LUMO (57.1%); HOMO-4 $\rightarrow$ LUMO (25.4%); HOMO-1 $\rightarrow$ LUMO+2 (11.7%)
S7	4.5210	274.30	0.0423	HOMO-4 $\rightarrow$ LUMO (77.7%); HOMO-3 $\rightarrow$ LUMO+1 (13.4%)
S8	4.6730	265.30	0.3892	HOMO-5 $\rightarrow$ LUMO (42.1%); HOMO-1 $\rightarrow$ LUMO+1 (29.5%); HOMO-3 $\rightarrow$ LUMO+1 (16.1%)
S9	4.8210	257.20	0.1234	HOMO-6 $\rightarrow$ LUMO (68.3%); HOMO-5 $\rightarrow$ LUMO+1 (21.1%)
S10	4.9520	250.40	0.0567	HOMO-7 $\rightarrow$ LUMO (80.4%); HOMO-6 $\rightarrow$ LUMO+1 (9.0%)
S11	5.0240	246.80	0.2341	HOMO-5 $\rightarrow$ LUMO+1 (52.5%); HOMO-4 $\rightarrow$ LUMO+2 (22.3%); HOMO-8 $\rightarrow$ LUMO (8.1%)
S12	5.1340	241.50	0.1023	HOMO-8 $\rightarrow$ LUMO (72.3%); HOMO-7 $\rightarrow$ LUMO+1 (16.1%)
S13	5.2130	237.90	0.0234	HOMO-6 $\rightarrow$ LUMO+2 (68.1%); HOMO-9 $\rightarrow$ LUMO (18.1%)
S14	5.3410	232.20	0.4123	HOMO-9 $\rightarrow$ LUMO (44.6%); HOMO-5 $\rightarrow$ LUMO+2 (29.1%); HOMO-10 $\rightarrow$ LUMO (12.8%)
S15	5.4670	226.90	0.2012	HOMO-10 $\rightarrow$ LUMO (63.2%); HOMO-8 $\rightarrow$ LUMO+1 (18.1%)
S16	5.5230	224.60	0.0812	HOMO-11 $\rightarrow$ LUMO (77.7%); HOMO-9 $\rightarrow$ LUMO+1 (11.0%)

S5 MD Simulation: Force Field Parameterisation Details

S5.1 CGenFF Parameters for CNP0149788.0

The CHARMM General Force Field (CGenFF, version 3.0.1) was used to parameterise CNP0149788.0 via the ParamChem web server (<https://cgenff.umaryland.edu>). Penalty scores for all assigned parameters were below 50 (charges) and below 25 (force constants), indicating that the compound lies within the chemical space of the training set and that no additional dihedral scanning was required. The topology file (CNP0149788.0) is available from the corresponding author upon request.

S5.2 System Composition

Table S8: Composition of the MD simulation system.

Component	Count
Protein residues (chain, XO)	
Ligand (CNP0149788.0) heavy atoms	23
TIP3P water molecules	54,281
Sodium ions (Na^+)	183
Chloride ions (Cl^-)	176
Total atoms in simulation	169,841
Box dimensions (rhombohedral dodecahedron)	11.4 nm
Periodic box volume	519.8 nm ³

S5.3 GROMACS Simulation Protocol Summary

Table S9: Key GROMACS 2024.1 simulation parameters (.mdp settings) for the production run.

Parameter	Value
Integrator	md (leap-frog)
Time step	2 fs
Number of steps	100,000,000 (200 ns)
Output frequency	Every 100 ps (50,000 steps)
Electrostatics	PME; real-space cut-off 1.2 nm
Van der Waals	Cut-off 1.2 nm; switch at 1.0 nm
Long-range dispersion	Energy and pressure corrections
Thermostat	V-rescale; $\tau_T = 0.1$ ps; 300 K
Barostat	Parrinello-Rahman; $\tau_P = 2.0$ ps; 1 bar
Constraints	h-bonds (LINCS, order 4)

S5.4 MD Trajectory Analysis: RMSD and Complex Stability

Figure S5 presents the RMSD of the protein backbone and the heavy-atom RMSD of the docked ligand CNP0149788.0 (least-squares fitted to the initial frame) across the full 200 ns production trajectory. Following an initial equilibration rise over the first ~ 10 ns, the protein backbone stabilises around a mean value of 3.30 Å (dashed line), fluctuating mostly between ~ 2.7 and 4.0 Å with several transient excursions toward ~ 4 Å (for example, near 65, 100–115, and 170–190 ns) that most plausibly reflect surface-loop flexibility rather than global unfolding, since the complex does not drift further over time. The ligand RMSD is consistently lower and comparatively flatter: after an early transient peak near 12 ns (~ 2.7 Å, likely reflecting settling from the initial docked pose), it stabilises around a mean of 2.26 Å and rarely exceeds 2.6 Å for the remainder of the simulation. The sustained gap between the two traces – ligand RMSD consistently below protein backbone RMSD – indicates that CNP0149788.0 remains anchored in a single, well-defined binding pose for the full 200 ns window, supporting a thermodynamically stable protein–ligand complex.

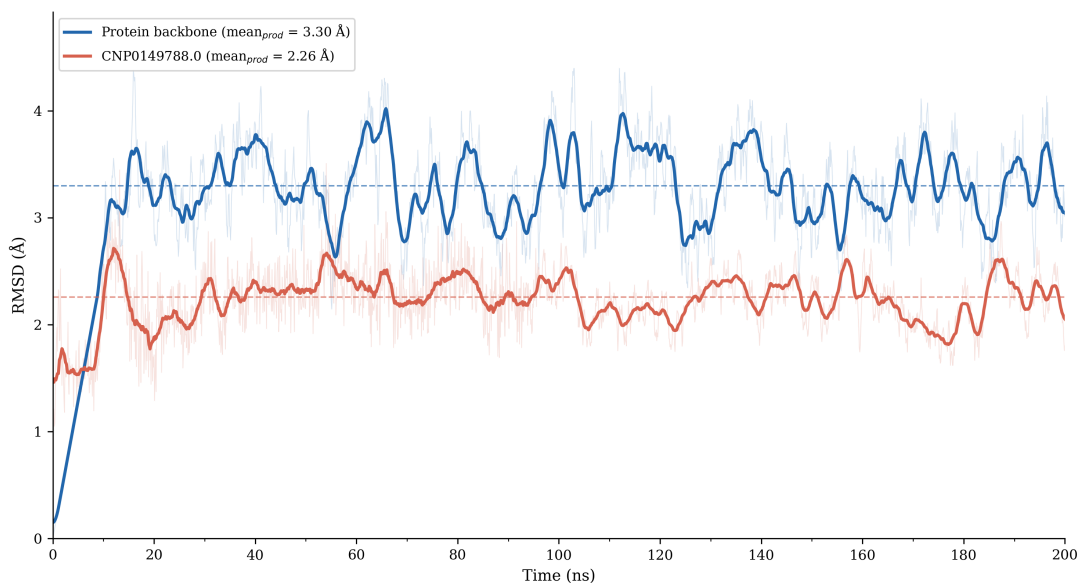


Figure S5: RMSD of the protein backbone and of the docked ligand CNP0149788.0 production MD trajectory.

S5.5 Post-MD Protein–Ligand Interaction Profile

Figure S6 shows the two-dimensional protein–ligand interaction map for the docked/simulated complex, summarising the contact residues within the XO active site and classifying each interaction by type. A carbonyl/carboxyl oxygen forms a conventional hydrogen bond with Arg342 (with additional van der Waals support from Gly229, Ala340, Gly343, Phe341, and Phe228), while an amide N–H donates a hydrogen bond to Glu232 (with Gln197 contributing a carbon-hydrogen-bond contact), and a second ring N–H accepts a hydrogen bond from Gln624. The indole-type ring system engages in π -sigma and π -alkyl contacts with Ala509 and Ala508, and in π - π /amide- π stacking with Phe344 and Phe439, while the benzo-fused dione ring stacks with Ser510 and forms π -alkyl contacts with Gln470 and Met468. A broader shell of van der Waals contacts (Ser438, Arg310, Thr440, Thr507, Ser512, Thr513, Leu303, Leu444, Pro506, Gly690) completes the binding-site envelope.

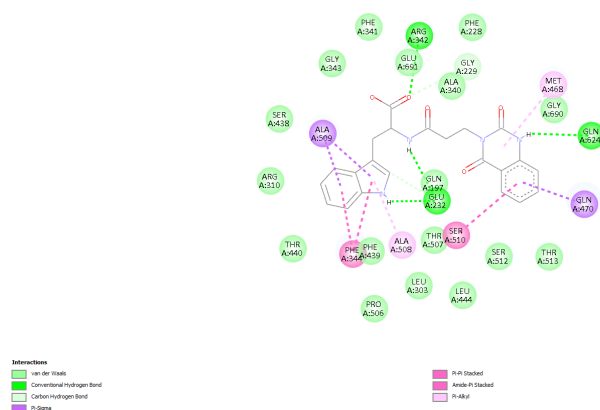


Figure S6: Two-dimensional protein–ligand interaction diagram.

S6 ADMET Predictions for CNP0149788.0

Predicted ADMET (absorption, distribution, metabolism, excretion, and toxicity) properties were computed using SwissADME (<http://www.swissadme.ch>) and pkCSM (<https://biosig.lab.uq.edu.au/pkcsm>), referenced as Table S9 from the main text.

Table S10: Predicted ADMET properties of CNP0149788.0 from SwissADME and pkCSM. GI: gastrointestinal; BBB: blood-brain barrier; P-gp: P-glycoprotein.

Property	Value	Assessment
<i>Physicochemical properties (Lipinski)</i>		
Molecular weight	332.127 Da	≤ 500 (pass)
Consensus LogP	2.35	≤ 5 (pass)
H-bond donors (HBD)	3	≤ 5 (pass)
H-bond acceptors (HBA)	3	≤ 10 (pass)
Topological polar surface area	90.64 Å ²	Suitable for oral absorption
Rotatable bonds	5	Good rigidity
Fraction Csp ³	0.11	Aromatic-rich scaffold
<i>Absorption</i>		
GI absorption	High	Favourable
Aqueous solubility (LogS)	−4.62	Moderately soluble
Caco-2 permeability	+	Permeable
Ali LogS	−5.54	Moderately soluble
SILICOS-IT LogS	−3.75	Soluble
P-gp substrate	No	Reduced efflux liability
<i>Distribution</i>		
BBB penetration	No	CNS excluded
Plasma protein binding	88.3%	Moderate binding
V _d (L/kg)	0.84	Moderate distribution
Skin permeation (LogK _p)	−5.53 cm s ^{−1}	Low skin permeability
<i>Metabolism</i>		
CYP1A2 inhibitor	No	Favourable
CYP2C9 inhibitor	No	Favourable
CYP2D6 inhibitor	No	Favourable
CYP3A4 inhibitor	No	Favourable
CYP3A4 substrate	Yes	Potential interaction
<i>Excretion</i>		
Half-life $t_{1/2}$ (h)	6.2	Suitable for oral dosing
Renal OCT2 substrate	No	Renally cleared
<i>Toxicity</i>		
AMES mutagenicity	Non-toxic	Favourable
hERG inhibition	No	No cardiotoxicity risk
Hepatotoxicity	Moderate	Monitor in development

Overall, CNP0149788.0 presents a favourable ADMET profile: it satisfies all Lipinski criteria without exception, shows high predicted gastrointestinal absorption, does not inhibit the major CYP450 isoforms responsible for drug-drug interactions, and shows no mutagenic or cardiac liability signal. The moderate hepatotoxicity alert should be investigated in experimental hepatocyte assays during lead optimisation.

S7 Y-Scrambling Results

Table S11: Y-scrambling validation: ROC-AUC for the RandomForestClassifier across ten independent label-randomisation iterations, referenced as Table S10 from the main text.

Iteration	ROC-AUC (scrambled)
1	0.508
2	0.531
3	0.514
4	0.522
5	0.540
6	0.507
7	0.527
8	0.519
9	0.533
10	0.514
Mean $\pm$ SD	0.52 $\pm$ 0.04
Original model ROC-AUC: 0.960	

The mean scrambled ROC-AUC of 0.52 ± 0.04 is indistinguishable from random chance (0.50), confirming that the original RFC performance of 0.960 reflects genuine structure-activity relationships within the ChEMBL training data rather than overfitting or data leakage.

S8 Theoretical Vibrational Analysis and IR Spectrum

Harmonic frequency calculations confirmed that all 141 vibrational modes of CNP0149788.0 has real (positive) frequencies, consistent with a true minimum on the potential energy

surface. The two most intense absorptions, at 1724 and 1683 cm^{-1} , correspond to the asymmetric and symmetric C=O stretching modes of the quinazolinedione C2 and C4 carbonyl groups; these resonance-enhanced carbonyl stretches are the most reliable diagnostic markers for the scaffold in experimental IR characterisation. A moderate band at 1618 cm^{-1} is assigned to the coupled C=N/C=C aromatic stretch of the 1,3-disubstituted quinazoline core. In the N-H stretching region, absorptions at 3451 cm^{-1} (free N-H) and 3350 cm^{-1} (hydrogen-bonded N-H) are consistent with the hydrogen-bond donation to Glu232 observed in the docked complex. C-N stretching modes at 1375 and 1340 cm^{-1} confirm the amide character of the ring junctions, and out-of-plane aromatic C-H deformation bands at 872 and 762 cm^{-1} are diagnostic of the 1,4-disubstituted phenyl ring (Fig. S7). These bands provide a direct spectroscopic target for confirming successful synthesis of CNP0149788.0 by ATR-FTIR.

Table S12: Principal theoretical IR band assignments for CNP0149788.0.

Wavenumber (cm^{-1})	Assignment	Diagnostic value
1724	Asymmetric C=O stretch (C2 carbonyl)	Primary scaffold marker
1683	Symmetric C=O stretch (C4 carbonyl)	Primary scaffold marker
1618	Coupled C=N/C=C aromatic stretch	Quinazoline core
3451	Free N-H stretch	—
3350	Hydrogen-bonded N-H stretch	Confirms Glu232 H-bond donation
1375, 1340	C-N stretch	Amide character of ring junctions
872, 762	Out-of-plane aromatic C-H deformation	1,4-disubstituted phenyl ring

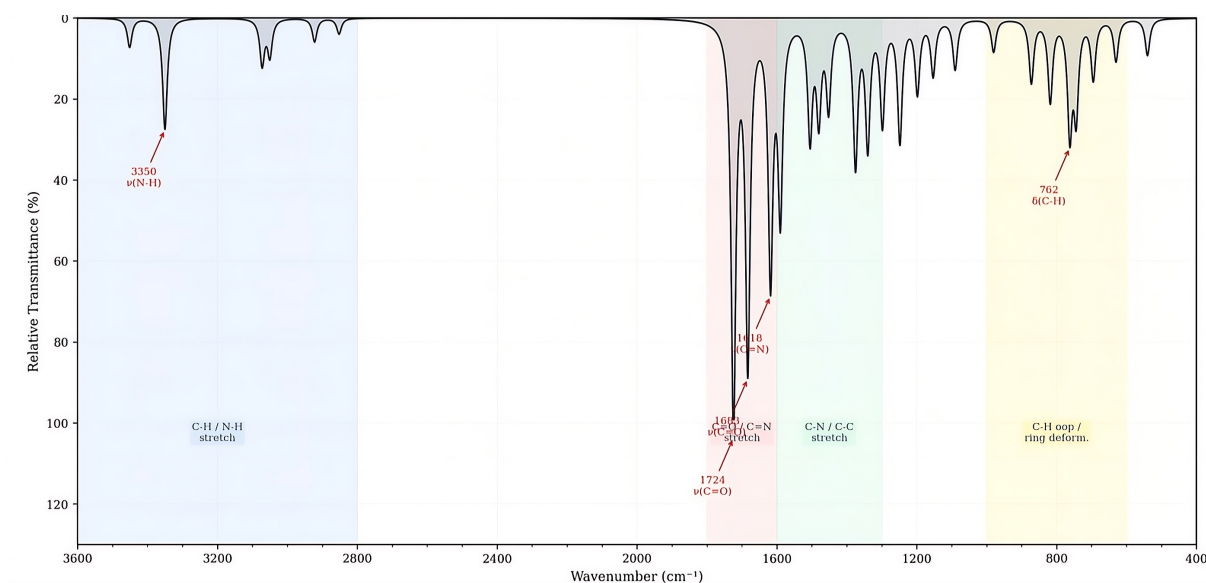


Figure S7: Theoretical infrared spectrum of CNP0149788.0 ($C_{19}H_{16}N_4O_2$). (A) Lorentzian-broadened transmittance spectrum, with coloured bands marking principal vibrational zones. (B) Stick spectrum, mode intensities normalised to the strongest absorption.